

A framework for general-purpose microscopic image analysis via self-supervised learning

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ABSTRACT

Combining materials science, artificial intelligence (AI) offers great potential for the extensive quantitative analysis and processing of material characterization associated with high-throughput experiments. However, due to the complex and diverse morphology of structural components, the development of AI models requires substantial annotation and models are usually task-specific with limited generalizability to different clinical applications. Here, we present a universal self-supervised learning framework for microscopic images. Our framework learns generalizable representations from unlabelled images and provides a pixel-wise segmentation for quantitative microstructure analysis in a variety of materials science applications. Specifically, the framework learns feature from a single image by means of self-supervised learning, and adapts it to a series of related tasks. We show that our method consistently outperforms several comparisons supervised or weakly supervised learning models in the context of various applications. Our approach provides a generalizable solution to improve model performance and alleviate the annotation workload of experts to enable practical AI applications from microscopic imaging.

1. Introduction

Microscopic analysis is of increasing importance in multiple domains, including nature, science, and technology [1–3]. The field is currently rapidly expanding, mainly facilitated by technological advances in the development of structure characterization and quantification techniques. The linkage between microstructure and material functional properties is a fundamental tenet of materials science and engineering [4,5], as the microstructure determines many physical properties, including mechanical performance, reactivity, thermal conductivity, and optical transparency [6,7]. Knowledge of the structure-property relationship enables the design and discovery of new materials, as well as the customization of their performance by controlling their structure during material processing and usage.

Structure quantification is the foundation for establishing the structure-property relationship. Over the years, high-resolution imaging techniques such as scanning electron microscopy (SEM), X-ray microtomography (X-ray μ CT), fluorescence microscopy (FM), and transmission electron microscopy (TEM) [8,9] have been commonly used to characterize microstructural information, followed by quantitative

analysis through image processing pipelines [10]. Microscopic segmentation, which assigns materials phases to each pixel or voxel in microscopic images, is essential for the precise quantitative analysis of specific parameters such as size and quantity. However, it remains a challenging task due to the complex and diverse morphology of structural components, as well as the low contrast and limited illumination in microscopic imaging systems [11,12].

Machine learning has been a long-standing research area that has led to significant achievements, ultimately transforming the paradigm of computer vision (CV) tasks. The introduction of Convolutional Neural Networks (CNNs) with a general architecture has brought substantial improvements to CV tasks [13]. With the recent surge in machine learning, state-of-the-art algorithms have been applied to microstructure systems, expanding the possibilities of traditional microstructure analysis [14–17]. Despite the valuable contributions of existing work to a wide range of microscopic segmentation tasks [18–27], practical applications are still limited by three main drawbacks. (1) High annotation costs: Currently, supervised learning demands accurate labeling, which can be expensive and time-consuming [28,29]. (2) Limited generalization ability: Existing approaches mostly deal with single-category image

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patterns and require model redesign when changing materials or imaging methods. Models lack the capacity to generalize to diverse complex tasks, indicating that models relying solely on supervised learning are not suitable for most real-world applications. (3) Interpretability remains relatively limited: Ensuring the safe and dependable translation of specific features learned during the training process into practical applications is a crucial step in this context.

Self-supervised learning (SSL) is a method within unsupervised learning that aims to address data inefficiency by deriving supervisory signals directly from data, without relying on expert knowledge in the form of labels [30–33]. Traditional SSL trains models to perform ‘pretext tasks’ for which labels are not required or can be automatically generated. This process leverages substantial amounts of unlabelled data to learn general-purpose feature representations that can readily adapt to more specific tasks. The combined strengths of strong generalization capabilities in these representations and the high performance achieved by self-supervised models in numerous downstream tasks highlight the significant potential of SSL in the context of materials characterization, especially when characterization tasks are diverse but labeled data is scarce [34].

In this work, we introduce a novel self-supervised learning (SSL)-based model for microscopic images and systematically assess its performance and adaptability in addressing various phase resolution tasks. Our core objective involves merging computer vision-based machine learning with neural network-based deep learning to enable precise microscopic segmentation without the need for manual annotations. Specifically, our model consists of two key components: (1) To enhance model interpretability and obviate the need for manual labeled data, we propose a machine learning unsupervised preliminary segmentation method to generate pseudo-labels. These pseudo-labels are fed back to the user before the neural network begins its learning process, offering

insights into the content the neural network is about to learn. (2) To maximize model accuracy and robustness, we employ a new loss function, namely, Smooth Cross-Entropy. This novel function reduces over-confidence in target categories, thereby enhancing the model’s generalization and robustness. Consequently, our novel approach ensures reliable microscopic segmentation decisions, extends the model to diverse complex tasks, and eliminates the need for annotations, thereby overcoming the aforementioned three drawbacks.

Fig. 1 provides an overall overview of this framework. We compare our model with multiple state-of-the-art supervised or weakly supervised learning models across various datasets, reporting both internal and external evaluation results. (1) We demonstrate the reliability of the pseudo-labels generated during the preprocessing stage, and our neural network is capable of rectifying labeling errors within these pseudo-labels. (2) We prove that our proposed model, trained with a single reference micrograph, can segment previously unseen micrographs taken from different parts or locations of the sample. (3) We establish the generality of our self-supervised microscopic segmentation approach across different material categories, structural components, and microscopic imaging modes. Its results are on par with, or even superior to, recently published supervised learning methods. Therefore, our framework holds the potential to expedite the paradigm shift in microscopic segmentation, moving from low throughput and manual processes to high throughput and automation.

2. Materials and methods

2.1. Self-supervised semantic segmentation

Self-supervised segmentation aims to classify pixels according to their similarities or characteristics, without the need for explicit

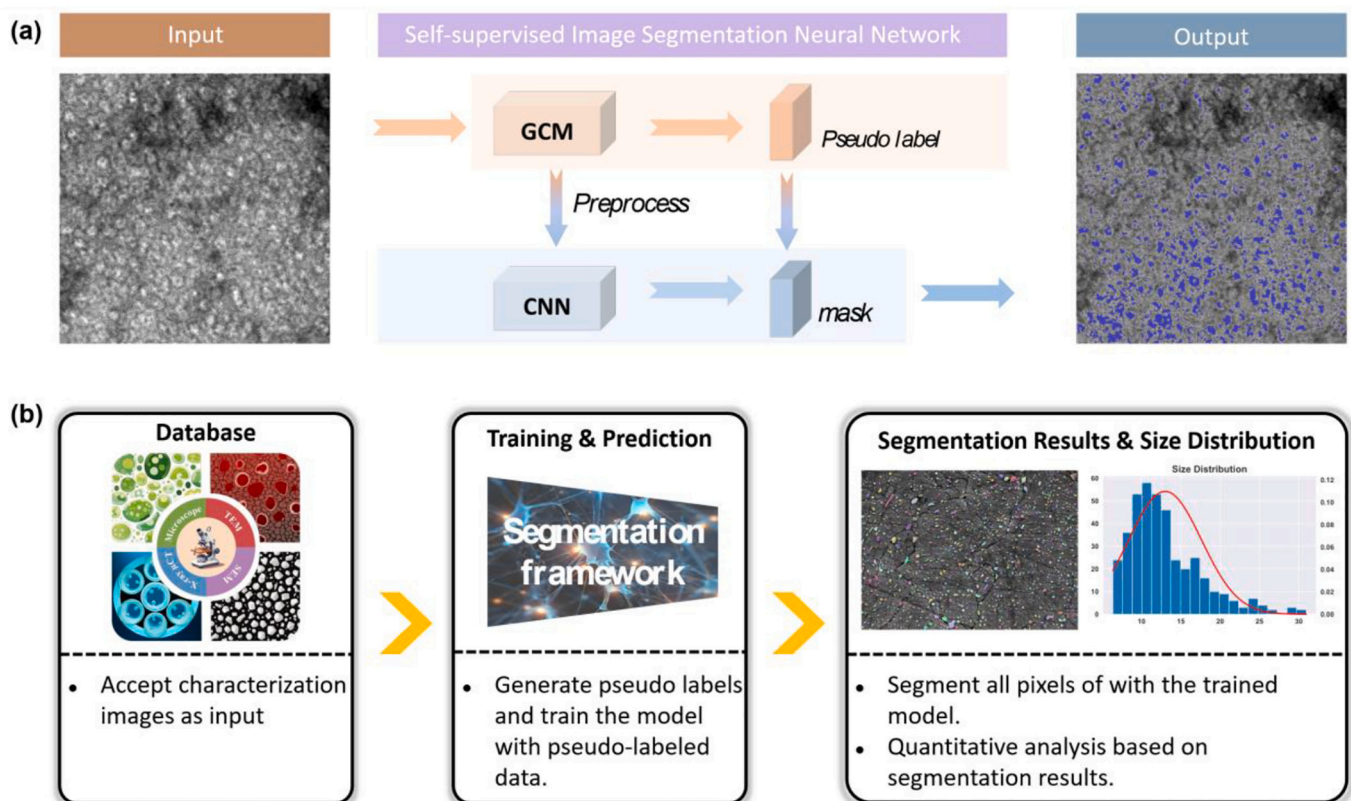


Fig. 1. Self-supervised learning framework for microscopic segmentation.

(a) We proposed the Generative Convergence Model (GCM) and integrated it with the k-means clustering algorithm to achieve pseudo labels based on the image features. We then used neural networks to learn and optimize the segmentation capabilities, (b) producing a versatile and robust machine learning model that can be applied to a variety of environments.

supervision during training. To achieve pixel-level precision in segmentation without relying on labels, we consider the following three constraints:

- **Spatial continuity:** Pixels that are close to each other should be assigned to the same cluster.
- **Feature similarity:** Pixels with similar features should be assigned to the same segment.
- **Model robustness:** The model should be robust to noise and variations in the input images.

To address these constraints, as shown in Fig. 1(a), we proposed the Generative Convergence Model (GCM) and integrated it with the k-means clustering algorithm to achieve pseudo labels based on the image features. In scenarios featuring relatively simple images, we prioritize the rapid processing speed of k-means to efficiently generate pseudo-labels. For these straightforward cases, k-means' swiftness and effectiveness make it the ideal choice.

However, for complex images, we rely on the powerful feature-extraction capabilities of the GCM model. This model's ability to handle intricate details and noise effectively allows it to generate more accurate pseudo-labels. By dynamically adapting our method based on image complexity, we achieve both efficiency and accuracy in pseudo-label generation.

We then used neural networks to learn and optimize the segmentation capabilities, producing a versatile and robust machine learning model that can be applied to a variety of environments (Fig. 1(b)).

2.1.1. Unsupervised machine learning for pseudo labels

We applied the GCM and K-means to accommodate different types of images. K-means is a simple and effective unsupervised learning algorithm for partitioning data points into K different clusters, where K is a user-defined parameter. In binary classification, K is typically set to 2 to separate the data into two clusters, representing the positive and negative classes. The mathematical representation of K-means is as follows:

$$\operatorname{argmin}_{i=1}^k \operatorname{dist}(x, \mu_i) \quad (1)$$

Where x represents the data point, μ_i denotes the centre of the i^{th} cluster, and $\operatorname{dist}(x, \mu_i)$ calculates the distance between the data point x and the i^{th} cluster center μ_i . In our binary classification problem, our goal is to assign data points to one of the two clusters, i.e., $i = 1$ or $i = 2$, in order to achieve the separation between positive and negative classes.

In scenarios where the foreground and background of images are more complex, we introduced the GCM (Generative Convergence Model), which is an iterative approach based on a predefined workflow for parameter selection. Compared to k-means, GCM requires more time, but offers a significant improvement in accuracy when dealing with images that are challenging to distinguish between foreground and background.

2.1.2. Neural network

For pixel-level classification, we use a convolutional neural network (CNN) as the core architecture of our model. CNNs first apply a series of convolutional layers to the input image to extract different features, such as edges, corners, and shapes. The output of the convolutional layers is then fed into a fully connected layer, which predicts the class of each pixel in the input image.

The networks have the following layers in order: a variable number of convolutional modules, each of which consists of a 3×3 convolutional layer with 64 filters, a rectified linear unit (ReLU) activation function, and a batch normalization layer. The output of the convolutional modules is then fed into a 1×1 convolutional layer, which reduces the dimensionality of the feature maps before they are fed into the fully connected layer. The structural information is listed in Table 1.

We chose to use a variable number of convolutional modules to allow

Table 1
Neural network structure.

Layer	Output shape	Kernel size	Channel size	Pad	Functions
Input	(input dim, H, W)	–	–	–	Conv2d, ReLU + batch norm
Hidden layers	(64, H, W)	3×3	64	1	Conv2d, ReLU + batch norm
Conv layer N	(k, H, W)	1×1	k	1	Conv2d, ReLU + batch norm
Output	(k, H, W)	–	–	–	Conv2d + batch norm

Input dim denotes the input dimension, H and W denote the height and width of the feature map, and k denotes the number of structural constituents' classes, which is 2 in this context.

the network to learn different levels of abstraction, which is beneficial for pixel-level classification.

Due to our utilization of traditional machine learning methods, we have generated some labels, transforming unsupervised learning into supervised learning. However, the challenge lies in the fact that the current labels are incomplete and contain errors. SmoothCrossEntropyLoss reduces the penalty on categories that are not the correct class but are somewhat similar, minimizing the impact of erroneous information in pseudo-labels. Its formula is as follows:

$$L = \frac{1}{N} \sum_{i=1}^N \left(- \sum_{j=1}^C (t_{ij} \bullet \log(p_{ij})) \right) \quad (2)$$

$$t_{ij} = \begin{cases} \frac{\text{smoothing}}{C-1} & j \neq \text{target} \\ 1 - \text{smoothing} & j = \text{target} \end{cases} \quad (3)$$

Where N is the number of samples, C is the number of classes. t_{ij} is the smoothed target for class j in sample i , p_{ij} is the predicted probability for classes j in sample i . target present for the true class, smoothing is a hyperparameter that controls the model's confidence in its predictions. It works by slightly reducing the assigned confidence for a correctly predicted class j and redistributing that reduced confidence evenly among the other classes.

Hyperparameters have a relatively minor impact on our model. We employ the Adam optimizer along with cosine annealing learning rate for 500 iterations, with a initial learning rate of 5×10^{-2} . cosine annealing learning rate can improve the training stability of the model, reduce oscillations, avoid getting stuck in local minima, and enhance training efficiency, learning rates are reduced smoothly. It is defined as follows:

$$\eta_t = \eta_{\max} \bullet \frac{1 + \cos\left(\pi \bullet \frac{t}{T}\right)}{2} \quad (4)$$

Where η_t is the learning rate at time step t . $\eta_{\max}$ is the initial learning rate. t is the current time step. T is the total number of training epochs. Cos is the cosine function.

The Xavier uniform initializer [35] is utilized as the kernel initializer. With the use of an NVIDIA Tesla P100-PCIE-16GB, the network training process takes approximately 5 min for a 768×768 micrograph, whereas evaluation only requires a few seconds.

2.2. Evaluation metrics

Common evaluation metrics in image segmentation include Intersection over Union (IoU), Precision, and Recall. IoU quantifies the overlap between the segmented image and its ground truth counterpart, serving as a key measure. Precision emphasizes the accuracy of positive predictions, measuring the ratio of correctly segmented pixels to all predicted positive pixels. Recall assesses the model's ability to capture

all relevant regions, measuring the ratio of correctly segmented pixels to all actual positive pixels.

$$IoU(A, B) = \frac{\|A \cap B\|}{\|A \cup B\|} \quad (5)$$

$$Precision = \frac{TP}{TP + FP} \quad (6)$$

$$Recall = \frac{TP}{TP + FN} \quad (7)$$

Where A and B are two segmentation masks for a given class, and TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative pixels, respectively.

2.3. Dataset

The helium bubble dataset was obtained by capturing the peak region under a focal distance of -500 nm using the Tecnai G2 F20 S-TWIN transmission electron microscope. The material used for the dataset was 304 L stainless steel, produced through Selective Laser Melting (SLM) and casting. The samples were irradiation with a flux of 350 keV He+

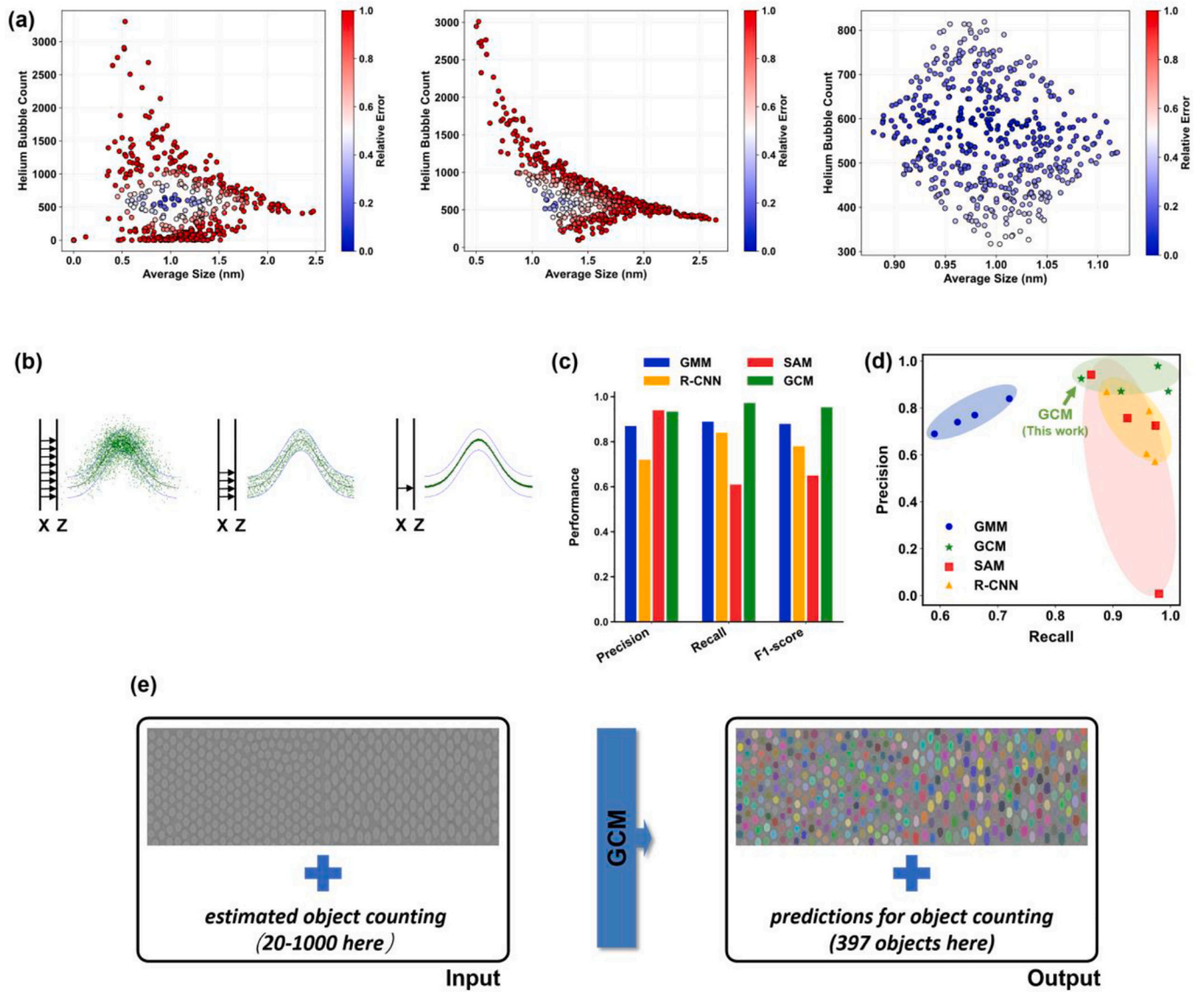


Fig. 2. Generative Convergence Model (GCM) for generating pseudo-labels.

(a) Accuracy of 500 predictions after each stage of GCM training, blue represents smaller errors, and red represents larger errors. GCM obtains better output segmentation results through a continuously converging range of input parameters. From left to right are the three stages of GCM parameter convergence: initial state, preliminary convergence state, and completion state, corresponding to the three stages in (b): the process of the model fitting the ground truth function. **Stage 1:** We first define the approximate range of the output results, and GCM randomly select parameters to generate a large number of predicted results, and check whether the results are within our preset range of results. **Stage 2:** According to the distribution of parameters whose results are within the range we preset, GCM extract their specific parameter ranges to filter out a small range of parameter distributions. **Stage 3:** GCM further converge the model, relying on the model's extraction of the image's own features, and further screen out the parameter groups that can successfully segment the image, and apply them as our results. **x** represents the parameter range to which the input belongs, and **z** represents the parameter range to which the output belongs. (c) Performance comparison graph between GCM and other models. (d) The Ashby-style plot of Recall and Precision of different machine learning models and generative convergence model (GCM) in this work. (e) A typical input and output of the GCM model. The input includes the raw image and a rough estimation of the number of objects of interest for analysis. The output consists of the model's predicted instance segmentation map and quantitative analysis based on that segmentation map. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ions at a density of 1×10^{17} ions/cm². In this irradiation process, the resulting damage per atom (dpa) was 2.35, and the helium concentration reached 5.5 atomic percent. The synthetic fiber dataset is freely available [22] and contains 537 X-ray μ CT micrographs with dimensions of 602×301 .

The fluorescence images of nuclei dataset is freely available [36] and consists of 24 SEM micrographs with spatial dimensions of 645×475 .

3. Results

In our self-supervised learning approach, we achieve comprehensive and clear microscopic segmentation without requiring any annotations. This is because the method relies solely on the inherent characteristics of the input images, resulting in relatively short training and inference times. We evaluated the versatility of our model in four different scenarios, representing typical segmentation challenges for various microscopic with differences in material categories, structures, compositions, and microscopic imaging methods: 1) SEM microphotographs of ultra-high carbon steel, 2) x-ray images of fiber cross-sections, 3) microscopic images of cells, and 4) TEM images of helium bubbles.

3.1. Generative convergence model

We evaluated GCM's image segmentation performance on the helium bubble dataset with 500 result sets at three stages: pre-training, post-training, and final state prediction. The experimental results are shown in 3 scatter plots in Fig. 2(a), where each point represents a model's prediction and the closer the point colour is to blue, the closer the result is to the ground truth. Fig. 2(c) and (d) show GCM's segmentation results after fine-tuning, compared to those of state-of-the-art supervised learning models and large language models for helium bubble detection. Quantitative results clearly demonstrate that GCM has potential in all metrics.

Fig. 3 shows the results of our pseudo-labelling, which vary in quality across datasets and have specific drawbacks. SEM images contain extra small noise, X-ray μ CT images have boundary fiber defects, Fluorescence microscopy (FM) images inaccurately detect certain cell shapes, and TEM images miss numerous tiny helium bubbles. In subsequent chapters, we will show the neural network's optimization results on pseudo-labels for each dataset and its robustness.

3.2. Model transfer-ability for SEM micrographs of ultrahigh carbon steel

Scanning electron microscopy (SEM) has become a fundamental analysis technique for studying a variety of materials [37]. In this paper, we explore the segmentation of spherical particles in SEM micrographs of ultra-high carbon steel (UHCS). Publicly available benchmark

datasets are available for UHCS segmentation from SEM micrographs [27].

Fig. 4 illustrates the segmentation results of carbon spherical particles, including the input image, overlay image, and size distribution between ground truth and predictions. The results demonstrate that our neural network can accurately locate and segment carbon spherical particles. Additionally, our model only requires a single image as input to train a robust model that can generalize well to other images in this task. The quantitative results in Table 2 show that our model consistently outperforms [24,27], which rely on supervised learning and weakly supervised learning, respectively.

3.3. Noise and low-resolution in X-ray μ CT micrographs of fiber

X-ray microtomography (X-ray μ CT) has emerged as a popular method for non-destructive three-dimensional (3D) imaging of fiber-reinforced composites (FRCs) at the constituent length scale [22]. This paper investigates fiber segmentation in fiber-reinforced composites (FRCs) in individual tomograms.

As shown in Fig. 5, the proposed method is compared to three standard methods: K-means clustering, OTSU method, and template matching. Irregularities in X-ray μ CT images, such as noise, unclear boundaries shapes, and missed edges, can be amplified in each image. K-means clustering and the OTSU method are widely used in pixel classification and foreground-background segmentation, but they can struggle with edge fiber detection and significant noise, leading to errors in noise misjudgement, especially in resin-rich areas. Template matching can only predict fixed sets of shapes and fails to capture the true contours of the fibers. Our self-supervised model addresses these challenges by learning from the majority of correctly segmented fibers, correcting pseudo-labels, treating each fiber as a whole, and focusing on boundary information. This enables the depiction of clear and complete fiber outlines.

3.4. Fluorescence micrographs of biological cells at different magnification and various types

Fluorescence microscopy (FM) techniques, widely used in biological research, enable the observation of cellular structures, histological features, protein distribution, and dynamic processes within biological samples. Additionally, FM imaging can be utilized for the study of cell labelling and gene expression [38]. The dataset consists of tumour tissue samples and bone marrow samples from neuroblastoma patients, serving as a benchmark for automatic cell nucleus segmentation experiments [18].

Fig. 6 shows the micro-segmentation of biological cells. These figures show input microscopic images, ground truth, and the segmentation

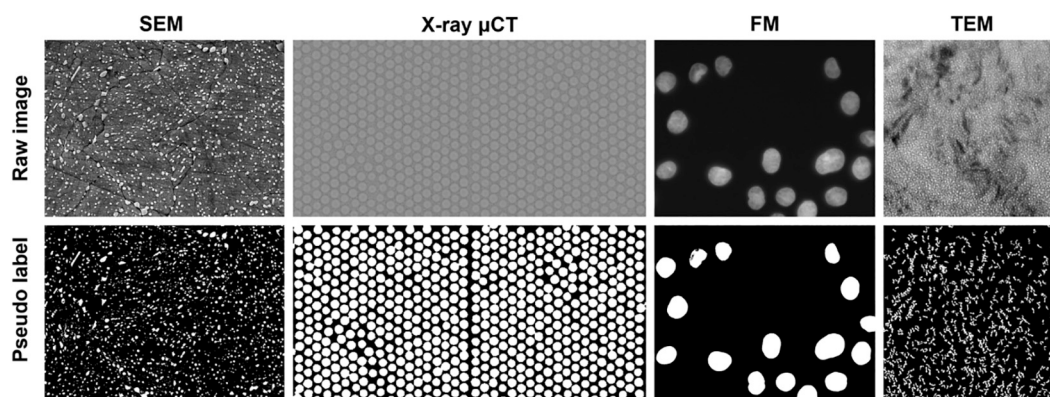


Fig. 3. Pseudo labels created using machine learning.

We integrate the GCM and K-means methods to directly generate pseudo-labels from input images. The first row is for input images, and the second row is for pseudo-labels.

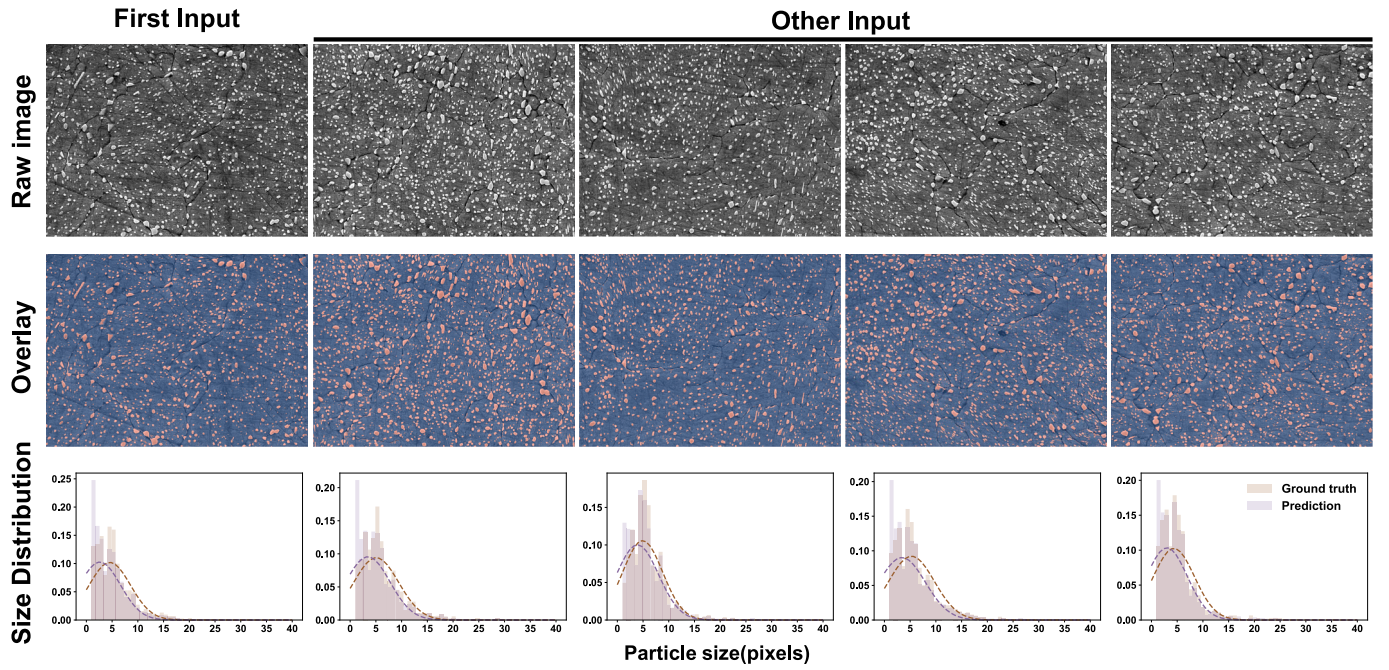


Fig. 4. Microstructure segmentation of spheroidite particles in ultrahigh carbon steels from SEM micrographs.

The first column contains training images, while the remaining columns show input images and the corresponding model predictions. The rows show the input images, overlaid model predictions, and a comparison of the predicted particle size distribution to the ground truth.

Table 2
Spheroidite particle segmentation performance.

	Training method	IOU	Precision	Recall
DeCost et al. [27]	Supervised	0.55	0.75	0.70
Juwon et al. [24]	Weak supervised	0.71	0.80	0.88
Ours	Self-supervised	0.705	0.830	0.824

Data are calculated across 24 SEM images.

overlay, with cells represented in colors and the background in black. We describe the segmentation of normal epidermal cells (HaCaT), followed by the segmentation of other cell types. The results show that our model successfully segments microscopic images into cells and backgrounds for all tasks, even without any manual annotations as supervision. Specifically, the model is robust to variations in resolution. When the magnification is reduced from $66\times$ to $10\times$, the model's segmentation accuracy is hardly affected. Furthermore, the model efficiently segments the Ganglioneuroblastoma and Neuroblastoma datasets. Table 3 shows the quantitative segmentation performance of our model on various tasks, using the metrics of precision, recall, and region intersection over union (IOU). Higher scores indicate better performance. Results shows that our model makes an improvement over the state-of-the-art pre-defined models [25].

3.5. Complex TEM micrographs of helium bubbles

Transmission electron microscopy (TEM) is the primary method for characterizing helium bubbles, due to its practical resolution limit of approximately 1 nm [39]. TEM allows for the observation of bubbles and voids, mainly through bright field imaging in thin regions of a foil, with changes in contrast revealing bubble size. In this paper, we explore the task of TEM image segmentation for nanoscale helium bubbles.

As shown in Fig. 7, TEM images of helium bubbles often contain a series of relevant defects, such as grain boundaries, dislocations, and non-uniform irradiation, which make helium bubble segmentation challenging. Traditional methods have had limited success in segmenting helium bubbles of 1 nm or smaller. We employed our model, which

takes a single input image, and tested it at different magnification levels and on helium bubble images from various 304 L stainless steel samples. The respective columns display segmentation images and overlay images. The results demonstrate that our model is less affected by image artifacts and can effectively differentiate helium bubbles (in blue) from the background (in black) even in complex imaging scenarios. This indicates that our model has learned spatial and feature relationships among pixels from pseudo-labels. The results in Table 4 suggest that our approach outperforms both supervised machine learning models [26] and deep learning models [19–21], showing significant improvements.

4. Discussions

4.1. GCM for pseudo-label generation

One of the key reasons why our method is applicable to various materials and characterization methods is that we introduced GCM and integrated it with unsupervised clustering algorithms. Which provides robustness against image noise, uneven illumination, and other imaging defects. By adjusting the region of interest through learning parameters, GCM overcomes the limitations of traditional image processing and machine learning methods in dealing with noise interference and capturing extremely small objects at the nanoscale. By adjusting the region of interest through learning parameters, GCM effectively denoises and balances the image using a mixture of self-learning parameter-based filters with different sizes and designs. The optimization of these parameters through learning from multiple filters holds the potential for achieving accurate results in a wide range of image detection tasks.

In contrast to supervised learning methods that require a large amount of annotated data and training processes, our unsupervised approach faces some common issues. For instance, the GCM model is based on fixed parameters for a single image, aiming to detect structural components or specific elements in the original image. As a result, retraining is necessary when new images exhibit significant variations in imaging conditions. This enables GCM to perform well in detection tasks but may lead to errors in segmentation tasks. To address these

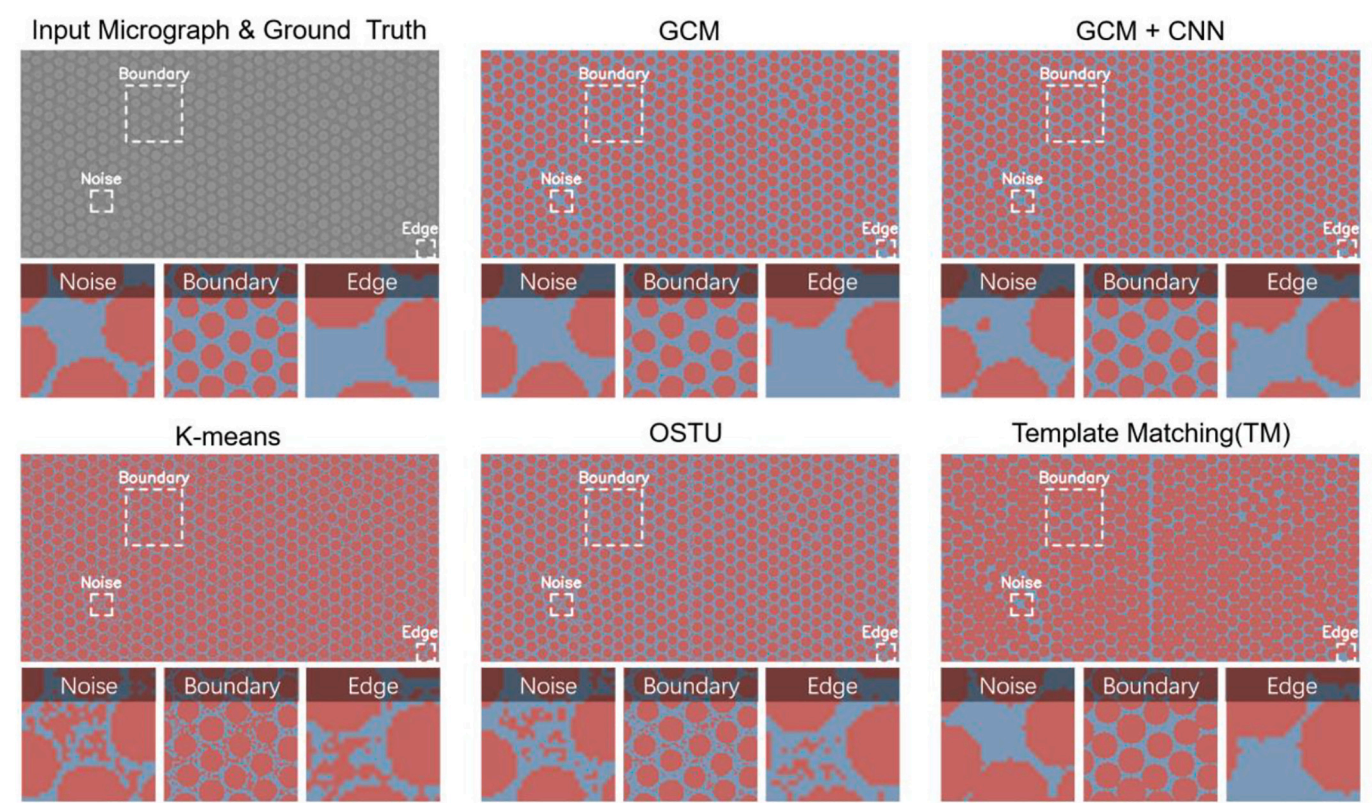


Fig. 5. Microstructure segmentation of fibers from X-ray μ CT micrographs. We compare our pseudo label (GCM) and model prediction (GCM + CNN) with several standard methods for fiber detection, including the K-means clustering, OTSU method, and template matching. The fibers are indicated in red, while the resins are indicated in blue. The white boxes indicate irregularities in X-ray μ CT images, such as noise, unclear boundaries, and missed edges, which are enlarged in each image below. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

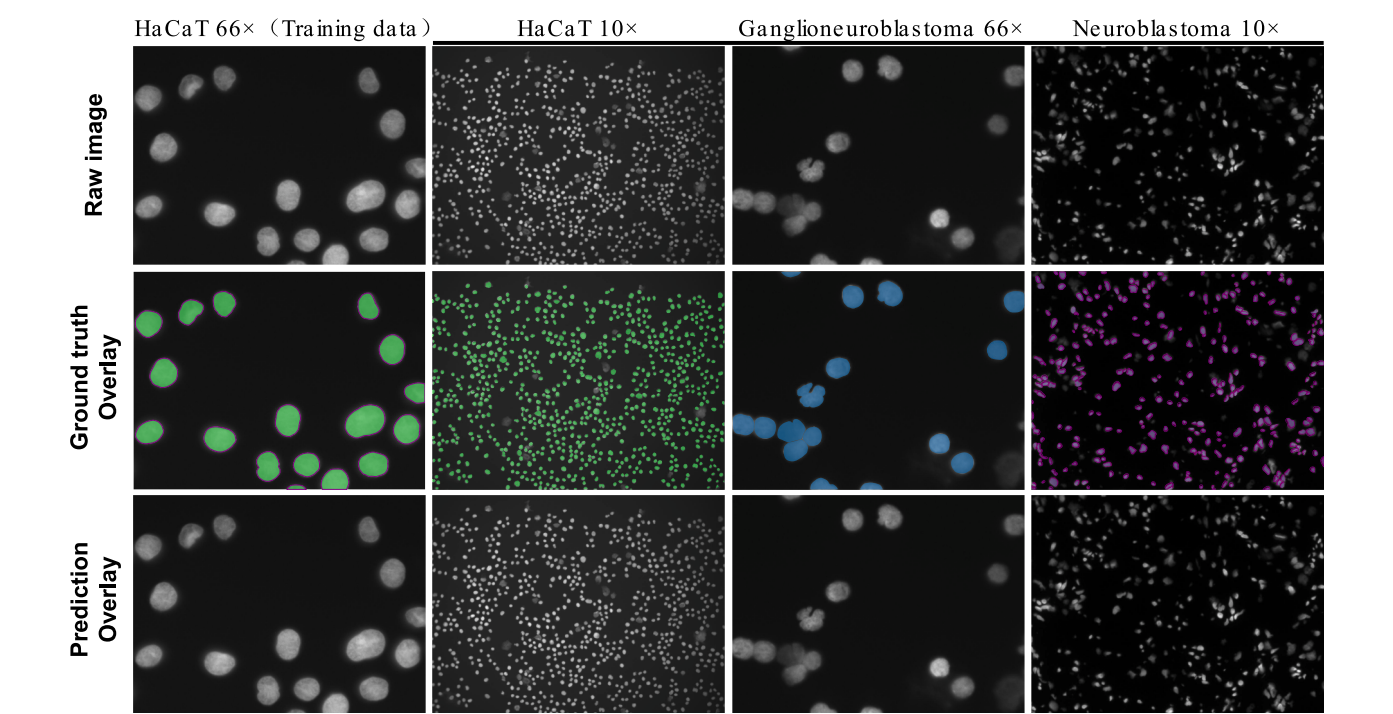


Fig. 6. Microscopic segmentation of biological cells from fluorescence microscopes. We evaluated our model's performance on fluorescence microscopy (FM) images at different magnification levels and under different intercellular conditions. The first column contains input images and their corresponding data. The remaining columns show FM images with abnormal cell morphologies or at different magnification levels. The first row represents the original training image, the second row is the ground truth, and the third row is the model's output.

Table 3
Biological cell segmentation performance.

	Training method	IOU	Precision	Recall
Giacopelli et al. [25]	Predefined model	0.842	0.854	0.986
Ours	Self-supervised	0.865	0.983	0.854

Data are calculated across 12 FM images.

limitations, we have adopted a different approach. Rather than directly predicting results, we utilize GCM in conjunction with unsupervised clustering to generate pseudo-labels. These pseudo-labels are then employed to train a neural network model, offering a solution that enhances the overall performance while mitigating the challenges associated with our unsupervised approach.

4.2. Neural network based microscopic segmentation

4.2.1. Smooth cross-entropy loss and model performance

Incorporating SmoothCrossEntropyLoss (SCELoss) enhanced model performance overall (Table B1), with the most significant improvement observed in the CNN model (IoU increase of 0.04). U²-net also demonstrated a modest improvement of 0.01, while U-net showed negligible change.

During training, we implemented early stopping to prevent overfitting and optimize computational resources, training was terminated if the loss did not decrease after 500 consecutive iterations. Analyzing the loss reduction curves from Fig. B1 reveals distinct training behaviors between models with and without SCELoss. While models employing SCELoss require longer training due to their gradual descent, they exhibit a smoother and more controlled process. This contrasts with models lacking SCELoss, particularly U-net and CNN, which experience

episodes of sudden loss spikes, indicative of Generative explosions that hinder further convergence. U²-net presents a mixed case, with its SCELoss implementation showing a slightly delayed loss reduction compared to its counterpart but no evident advantage in terms of smoothness.

The IoU trajectories over iterations (Fig. B2) demonstrate a common pattern for all models: a peak in IoU followed by a gradual decline. Notably, both CNN models achieve their maximum IoU almost synchronously. However, the model utilizing SCELoss exhibits superior stability in the later training stages, resulting in a higher final IoU compared to the standard CNN model. The U-net models, with and without SCELoss, demonstrate comparable performance curves. While the U²-net model with SCELoss requires more iterations to attain its peak IoU, its ultimate performance converges with its counterpart.

SCELoss incorporates a smoothing mechanism, which involves applying a smoothing treatment to the target labels when calculating the model's loss. This mechanism enhances the robustness of the model during training and provides a certain alleviation of issues such as overfitting. For simpler models like CNNs, this smoothing effectively prevents the model from making overly confident predictions that could

Table 4
Helium bubble detection performance.

	Training method	F1-score	Precision	Recall
Wu et al. [26]	Supervised	0.799	0.694	0.940
Anderson et al. [21]	Supervised	0.78	0.84	0.72
Chen et al. [20]	Supervised	0.77	0.75	0.88
Sainju et al. [19]	Supervised	0.794	0.780	0.810
Ours	Self-supervised	0.83	0.87	0.79

Our data are calculated across 6 TEM images.

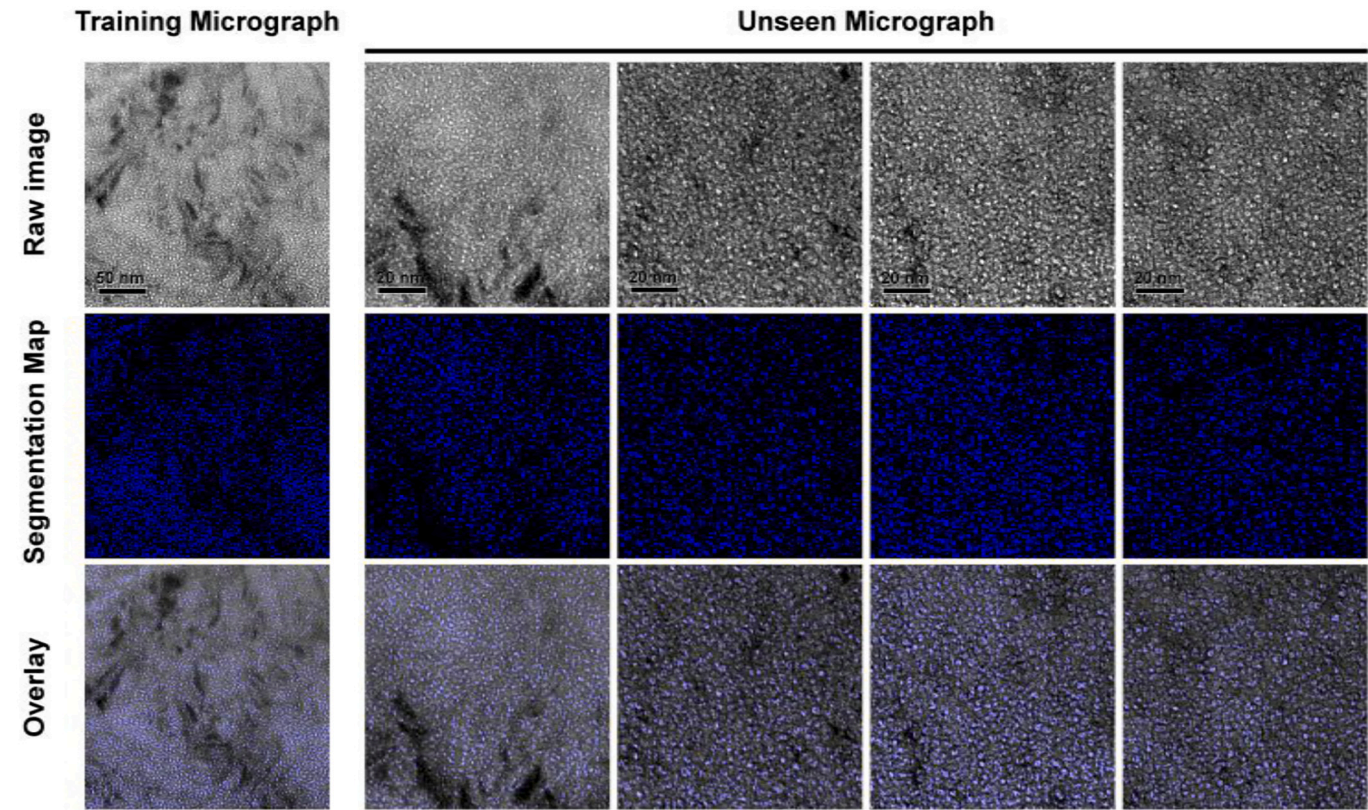


Fig. 7. Microscopic segmentation of helium bubbles from TEM. We evaluated our model's performance on TEM microscopy images at various magnification levels with different 304 L stainless steel samples. The first and second columns represent 304 L stainless steel samples manufactured through Selective Laser Melting (SLM) at different magnification levels, while the remaining columns represent cast 304 L stainless steel samples.

lead to prediction errors. On the other hand, for more complex models such as U-net, SCELoss can, to some extent, mitigate the impact of gradient explosions, resulting in a smoother training process. Additionally, since U²net is equipped with more complex structures, including residual blocks, to control convergence, the influence of SCELoss appears relatively minor.

These results validate our initial hypothesis that SmoothCrossEntropyLoss (SCELoss) would prevent class bias and improve accuracy. Notably, the CNN with SCELoss achieves comparable or even superior IoU performance to more complex models, and with significantly faster training speeds. This observed acceleration is surprising, considering that the SCELoss CNN requires nearly twice the iterations to reach stability as its counterpart. However, this increased iteration count translates to only a few seconds of additional training time. We believe this unexpected efficiency arises from the regularization effect of SCELoss. The class smoothing applied during training likely brings the model closer to optimal solutions with each iteration, streamlining the loss calculation and backpropagation process, and ultimately leading to faster convergence.

While SCELoss necessitates more iterations for convergence, we posit that this characteristic is not inherently detrimental due to two key aspects:

1. **Gradient Explosion Mitigation:** The early convergence observed in CNN and U-net models without SCELoss is often associated with gradient explosions, leading to premature loss reduction that can hinder optimal performance. Conversely, SCELoss, while converging more gradually, promotes a more sustained and controlled loss descent, ultimately contributing to enhanced model accuracy.
2. **Negligible Impact on Lightweight Models:** Notably, both CNN and U-net models, regardless of SCELoss implementation, attain their IoU peaks nearly simultaneously during training (Fig. B2(b)). This suggests that the use of SCELoss within relatively lightweight architectures does not impose a significant increase in overall training cost.

Looking beyond the current dataset, we envision SCELoss holding even greater potential when applied to larger datasets. While our models encountered overfitting challenges, artificially restricting the number of training iterations is often impractical in real-world scenarios. This is particularly true when accurate annotations for test subjects are scarce, necessitating the common practice of training models until stable convergence is achieved. As datasets expand in size, the ability of SCELoss to prevent gradient explosions and optimize the training process becomes increasingly valuable, enabling models to better adapt to large-scale data environments. We hypothesize that once the overfitting issue is mitigated through a sufficiently large dataset, SCELoss can initiate a more effective and efficient training process, leading to enhanced performance outcomes.

Therefore, based on the observed capability of the CNN model with SCELoss to deliver comparable performance to more complex architectures while maintaining faster training speeds, we judiciously adopt the CNN model as the foundation for our approach.

4.2.2. Combinational of GCM and CNN

To enhance the accuracy of our function while maintaining its capability on data outside the training dataset, we introduce neural networks to learn from pseudo-labels. Artificial neural networks, a fundamental component of deep learning, have demonstrated remarkable potential in image processing, revolutionizing the field of computer vision [12]. However, current neural network training heavily relies on a substantial amount of annotated data and training processes.

This dependence on manual annotation leads to high costs associated with image acquisition and annotation. We attribute this to two primary factors: (1) Neural networks require manually annotated data for learning, which can be time-consuming and resource-intensive to obtain. (2) The design of neural networks necessitates sophisticated

training with diverse data to address various challenges and account for potential errors in annotations. This becomes particularly challenging when data availability and computational resources are limited. Consequently, we propose a novel approach to tackle this issue.

Our self-supervised learning framework leverages GCM to provide the necessary training data for neural networks. Acknowledging that the data generated by GCM may have imperfections, we design the SCELoss function to prevent the model from becoming overly confident. This mitigates the impact of imperfections in the training data and addresses the issue of imprecise annotations. Additionally, our model is intentionally designed to be smaller, with the goal of training it once to address a specific problem. This approach allows each individual model to have a narrower scope, leading to improved training speed and accuracy. By leveraging SCELoss, the CNN achieves training accuracy comparable to more complex models while significantly accelerating the training process. Furthermore, our approach significantly reduces overall training time and the duration required to produce image segmentation results compared to the extensive data-driven training of large-scale models designed for board applicability. Notably, our model outperforms existing state-of-the-art methods across various individual tasks.

5. Conclusions

Efficient and accurate analysis of microstructures in materials science has driven the foundation of this research. We address the problem of binary microscopic segmentation through a novel self-supervised learning approach, focusing on (1) overcoming the limitations of annotations by leveraging image-based features for segmentation, (2) obtaining more precise pseudo-labels, and (3) enabling neural networks to learn more accurate information by avoiding errors in pseudo-labels.

Specifically, we make the following contributions: First, we utilize machine learning to generate pseudo-labels, transforming supervised learning into self-supervised learning, eliminating the need for manual annotation. Second, we introduce a new loss function to reduce the impact of incorrect labels on our neural network. Third, our model, trained on a single reference image, generalizes well to a wide range of related images. Fourth, we conduct comprehensive experiments under various image conditions to demonstrate the broad applicability of our model and its robustness to image disturbances such as noise, clarity, magnification differences, and uneven lighting. Our model achieves comparable or superior accuracy to supervised or weakly supervised learning methods, without requiring any human labeling effort. To the best of our knowledge, our method is the first general-purpose microscopic segmentation method that achieves high accuracy on various types of data without requiring manual labels.

Overall, our framework shows promising practical applications in various fields, such as nuclear materials science, life science as well as other disciplines that require accurate and efficient analysis of materials at the microscopic scale. This study demonstrates the potential of machine learning to enhance and accelerate material science research by enabling accurate and efficient characterization and analysis of microscopic materials properties. This opens doors to accelerated and data-efficient techniques that may transform the development of materials with improved properties and performance.

Author contributions

Z.Z., X.Y. designed the study. Z.Z. collected the data sets and conducted the experiments throughout the project and wrote the manuscript. J.W. revision of the manuscript. All the authors participated in the discussion of the results.

CRedit authorship contribution statement

Zhiwei Zheng: Conceptualization, Data curation, Formal analysis,

Investigation, Methodology, Software, Validation, Writing – original draft, Writing – review & editing. **Xuezheng Yue:** Funding acquisition, Investigation, Project administration, Writing – original draft, Writing – review & editing. **Jincheng Wang:** Formal analysis, Supervision, Writing – original draft, Writing – review & editing. **Juan Hou:** Data curation, Resources, Supervision.

Declaration of competing interest

The authors declare no competing interests.

Appendix A

Model Architecture and Description of Generative Convergence Model (GCM).

This section provides a detailed model architecture and formula explanations for the Generative Convergence Model (GCM) mentioned in Section 2.1.1 and Section 3.1 of this article.

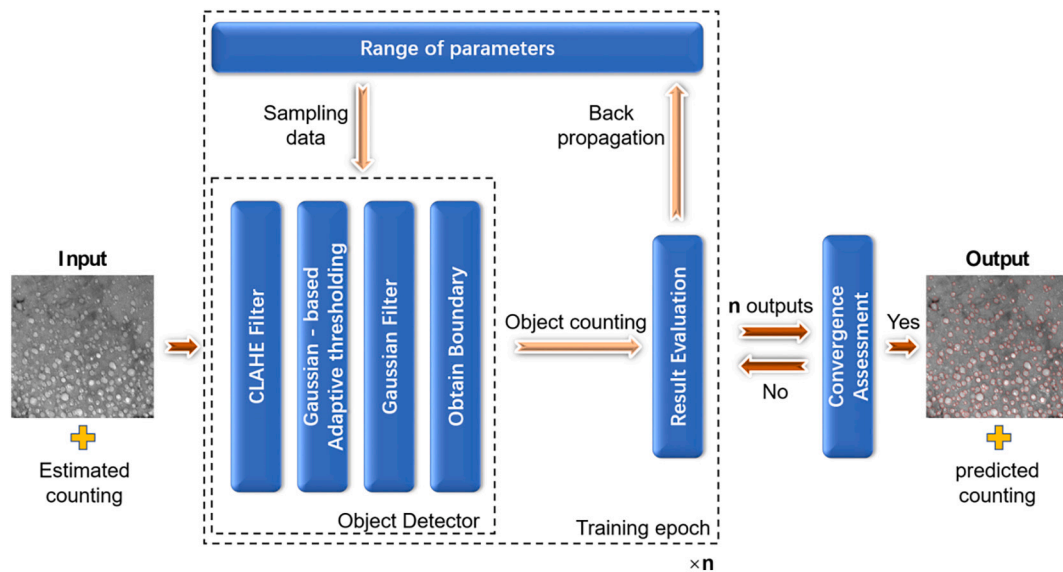


Fig. A1. Overall training process of the GCM model and an example input and output image of a helium bubble.

As shown in Fig. A1, the GCM consists of an object detection process and a training process. We input the object detector with different sets of parameters sampled each time, along with the image itself, and obtain a counted detection result. Subsequently, we evaluate whether the result falls within our initial estimation range and perform backpropagation based on the distance between the result and the initial estimated data to update our parameter sets.

Each filter in the Python-OpenCV library serves a specific purpose. Here are the descriptions of some used filters along with the convergence assessment loss formula:

CLAHE Filter (Contrast Limited Adaptive Histogram Equalization): CLAHE is an adaptive histogram equalization method that enhances the contrast of an image by redistributing pixel intensities. It divides the image into small regions called tiles and applies histogram equalization independently to each tile. This filter is particularly useful in enhancing local contrast in images.

Gauss-based Adaptive Thresholding: Gauss-based adaptive thresholding is a technique used for image binarization. It calculates a local adaptive threshold for each pixel based on the mean and standard deviation of the pixel intensities in its neighborhood. Pixels with intensities above the threshold are set to one, while those below are set to zero.

Gaussian Filter: The Gaussian filter is a widely used image smoothing filter. It convolves the image with a Gaussian kernel, which applies a weighted average to the pixels. The weights are determined by the Gaussian distribution, giving more importance to nearby pixels. This filter effectively reduces noise and blurs the image.

We have defined a simple convergence assessment loss formula:

$$L_{con} = \frac{NS}{PS + NS} \quad (A1)$$

Where NS (negative samples) represents the number of negative samples predicted by the model, and PS (positive samples) represents the number of positive samples predicted by the model. The model outputs a fixed total number of samples during training. The loss gradually decreases as the number of positive samples increases, and the model is trained in the direction of minimizing the loss.

To address the issue of uneven brightness in the entire image, we first applied CLAHE to partially balance the overall brightness and reduce interference in image segmentation. This serves as a form of regularization, bringing the grayscale values of all pixels to a more evenly distributed range, making it easier to identify regions of interest since they often have distinct grayscale characteristics. Next, we used Gauss-based Adaptive

Thresholding as a replacement for traditional fixed thresholding filters. Gauss-based Adaptive Thresholding is more sensitive to the target and its surrounding pixels, making it easier to segment images in complex environments. Fixed thresholding filters like Otsu tend to produce large areas of positive or negative labels when segmenting small objects. This is because fixed thresholding is based on global rather than local information, resulting in inadequate processing of local details. Lastly, we employed Gaussian filtering to remove noise and mitigate misclassifications in subtle areas. Since all our filter parameters were learned during training, our model exhibits greater flexibility in adapting to various scenarios, providing improved applicability and accuracy compared to preset parameters.

Appendix B

This section presents the ablation experiment results comparing our model with several widely-used baseline image segmentation models: U-net [40] and U²-net [41]. The experiment utilized the annotated Microstructure segmentation of spheroidite particles in ultrahigh carbon steels dataset [27]. The aim is to demonstrate that our model offers significant time savings during the training process while maintaining a minimal decrease in segmentation accuracy.

Table B1

The time required to train until model convergence and evaluation results.

Neural network	Training time (s)	Iterations	IoU
U-net	181±9	915±45	0.705±0.001
U-net + SCELoss	573±33	3000	0.705±0.001
U ² -net	892±7	3000	0.704±0.001
U ² -net + SCELoss	808±12	3000	0.705±0.001
CNN	36±11	1135±335	0.66±0.04
CNN + SCELoss	42±2	2215±105	0.70±0.01

SCELoss means Smooth Cross-Entropy Loss here. Models have a learning rate of 0.05 and use SCELoss with a smoothing parameter of 0.1. Training is stopped for each model if there is no loss decrease after 500 iterations. We conducted three groups of experiments for each case and give the statistical results.

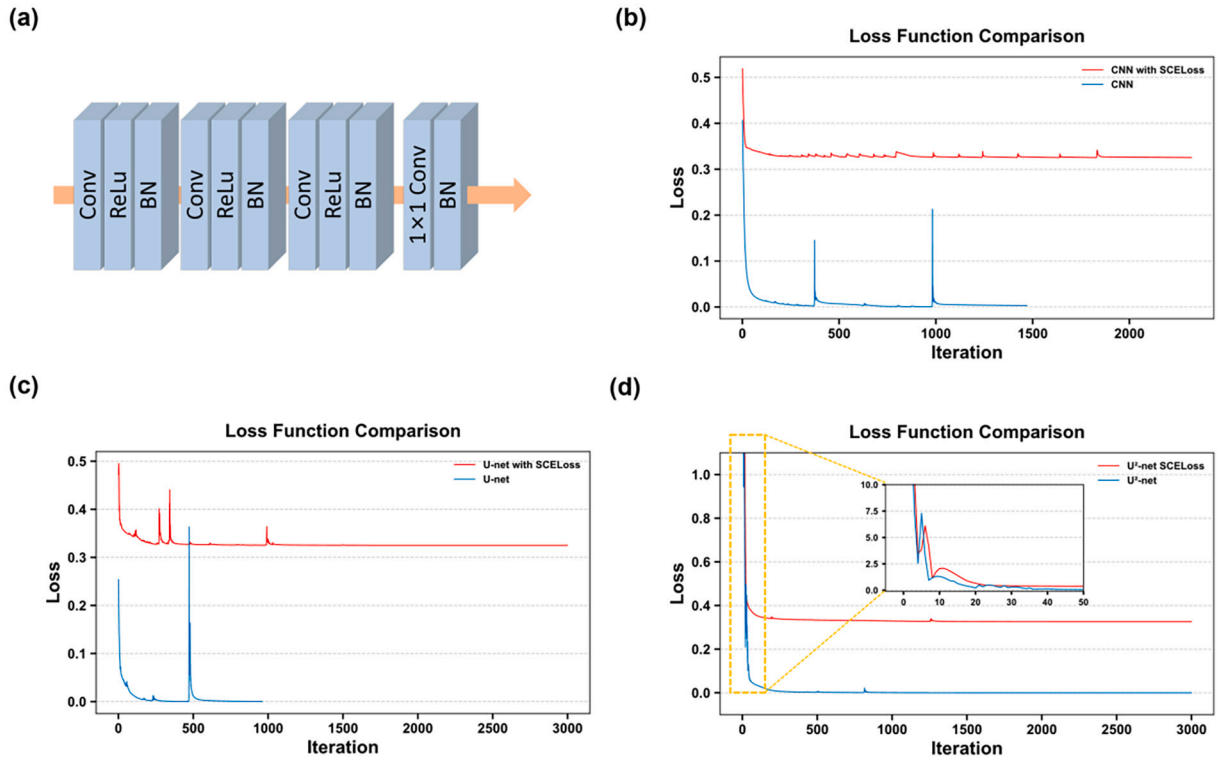


Fig. B1. The CNN model used in this paper and the decreasing trend of the Loss function in the ablation experiments with iterations.

The loss of models using SCELoss will converge to a value related to the smoothing parameter, rather than reaching 0. In these models, the smoothing parameter is set to 0.1. (a) The CNN model used in this paper consists of three convolutional layers composed of Convolution, ReLU, and Batch Normalization (BN) to extract features. It is then followed by a final 1×1 convolutional layer and Batch Normalization to provide pixel classification results. (b) The loss decreasing trend for the CNN model, (c) U-net model, and (d) U²-net model is shown. We employed early stopping during training, where if the loss does not decrease within 500 iterations, the training is terminated prematurely.

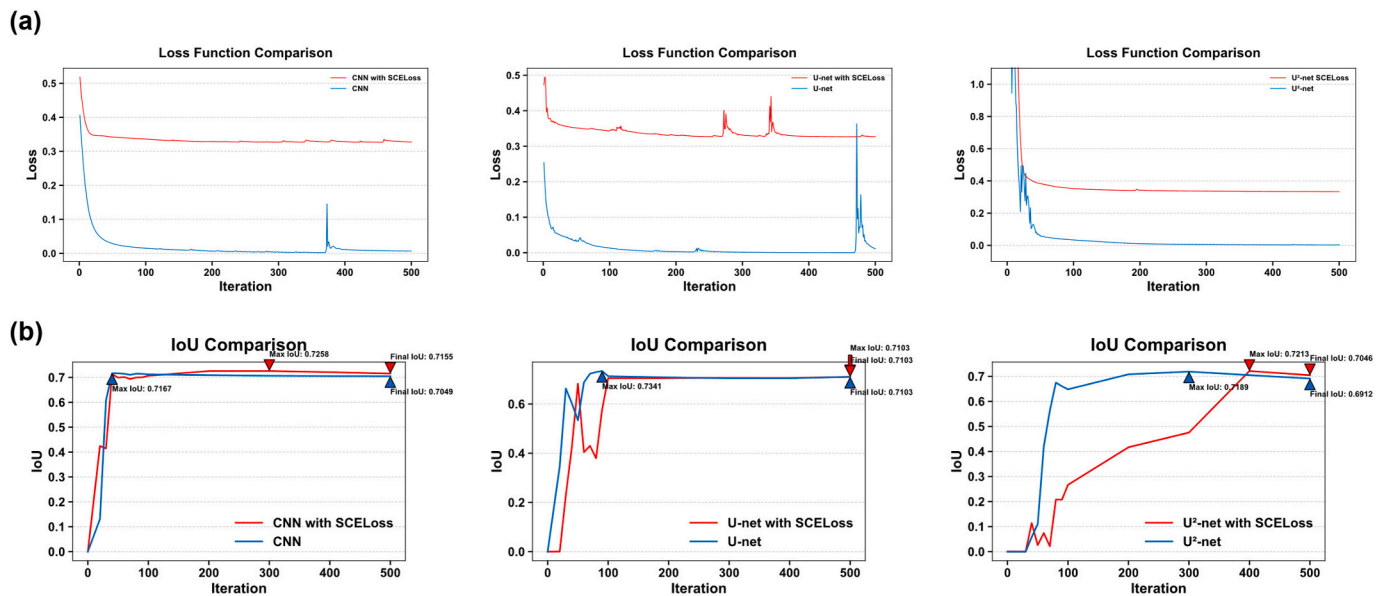


Fig. B2. Loss and IoU for the first 500 iterations of different models.

(a) From left to right: Loss for CNN, U-net, and U^2 -net. (b) From left to right: IoU for CNN, U-net, and U^2 -net. With pseudo label for training, the IoU is 0.68, and the CNN with SCELoss has a 1.4 times improvement over that without it (0.0355 and 0.0249, respectively) in the 500th iteration. The U-net with SCELoss is in a steady rise after 100 iterations, while the IoU of the general U-net has a slight decline in the later training period. The U^2 -net slows down the convergence with SCELoss, but the IoU is improved. We speculate that there is some functional duplication between SCELoss and the residual module of U^2 -net itself, resulting in slower training.

Appendix C. Supplementary data

Data availability: The SEM image data and Python code used in this study can be found online at <https://doi.org/10.1016/j.matchar.2024.114003>. For more information, please email: usst-yzyz@usst.edu.cn.

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